



Cairo University
 Faculty of Engineering
 Credit Hours System
Communications and Computer Engineering Department

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Course Code EECS 304
 Course Title Control I
 Course Instructors Dr. Ragia Badr - Dr. Nasr Antar
 Course TA Eng. Hassan El-Menier

Sheet 5: Routh Stability Criterion

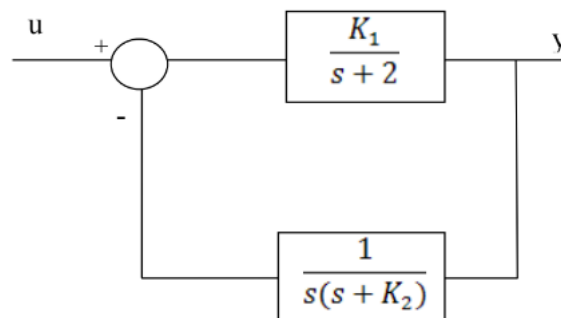
Question (1):

Test the stability for the following polynomials; hence determine whether any of the roots are in the RHP (Right Half Plane):

1. $F(s) = s^5 + 3s^4 + 2s^3 + 6s^2 + 6s + 9$
2. $F(s) = s^5 + 5s^4 + 11s^3 + 23s^2 + 28s + 12$

Question (2):

For the system shown in figure, find the conditions on K_1 and K_2 to make the system stable. Plot the region of stability for K_1 and K_2 .



Question (3):

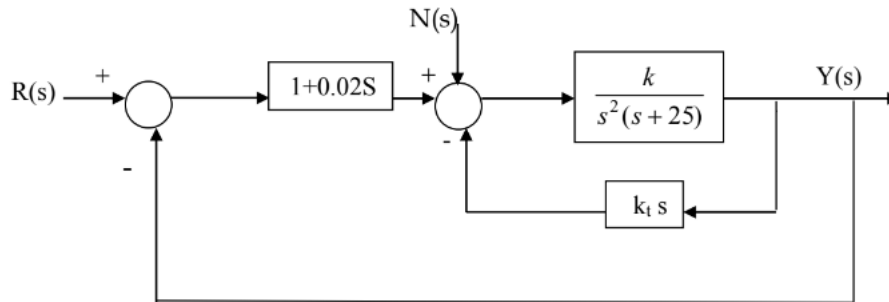
A unity feedback control system has the forward transfer function:

$$G_p(s) = \frac{K_v}{s(4s+1)(s+1)}$$

1. The steady state value of the error is desired to be less than or equal to 0.1 for a reference input $r(t) = 1 + t$. Determine the minimum value of K_v that satisfies this requirement.
2. Check the stability of the system for the value of K_v obtained in part (1) and comment on your result.

Question (4):

The block diagram of a feedback control system is shown in the figure. The error signal is defined to be $e(t)$. Find the steady state error of the system in terms of K and K_t when the input is a unit ramp function. Give the constraints on the values of K and K_t so that the answer is valid. Let $n(t) = 0$ for this part.

**Question (5):**

The block diagram of a control system is shown the following figure. Find the step, ramp and parabolic error constants. The error signal is defined to be $e(t)$. Find the steady state errors of the system in terms of K and K_t , when the following inputs are applied:

1. $r(t) = 6 + 8t$
2. $r(t) = 2t + 7t^2$

What constraint must be made on the values of K and K_t so that the answers are valid? Determine the minimum steady state error that can be achieved with a unit ramp input.

